

Surface Chemistry

Theory: Colloid is a state of a substance (matter). A colloid is a heterogeneous system in which one substance with particle size 1 nm to 1000 nm, dispersed in another substance called dispersion medium. So colloid is not a substance but a particular state of a substance which depends on size of particles. Colloids do not pass through animal membrane or parchment paper.

(Note: $1\text{ nm} = 1 \times 10^{-9}\text{ m} = 1 \times 10^{-7}\text{ cm}$)

Colloidal system consists of two phases :

- a. Disperse phase :** It is discontinuous phase of colloidal system having particles of colloidal size (1 nm to 1000 nm) dispersed in the dispersion medium.
- b. Dispersion medium :** The medium in which colloidal particles are dispersed is called dispersion medium. It is a continuous phase of a colloidal system.

Colloidal solutions (sols) are classified into two types :

- a. Lyophilic sols :** In this type of colloidal solutions (sols), the disperse phase has strong affinity for the dispersion medium. These are reversible sol as the solid obtained after evaporation of liquid may be reconverted to the sol by simply shaking with the dispersion medium.

Examples: Starch sol, gum sol etc.

- b. Lyophobic sols :** In this type of colloidal solutions (sols), the disperse phase has little or no affinity for the dispersion medium. These are irreversible sol as their precipitated mass cannot be reconverted to the sol by simply shaking it with the dispersion medium.

Example: Ferric hydroxide sol, Aluminum hydroxide sol etc.

Date: / /

Experiment No. 10

Aim: To prepare lyophilic sol (colloid) of starch.

Apparatus: Two beakers (50 mL, 250 mL), glass rod, funnel, filter paper, tripod stand, wire gauze, burner, etc.

Chemicals: Distilled water, soluble starch (5 g).

Theory: The sols which directly formed simply by mixing substances such as starch, gum, gelatin, rubber, etc. in liquid dispersion medium (water) are called lyophilic sols.

Starch form a lyophilic sol, when water is used as dispersion medium. The starch sol can be prepared by heating starch and water at about 100°C . It is quite stable and is not affected by the presence of any electrolytic impurity.

Procedure:

1. Take 5 g of starch in a 50 mL beaker and add few drops of distilled water to it and make a thin paste of starch.
2. Take 100 mL distilled water in a 250 mL beaker and heat water in a beaker till it starts boiling.
3. Pour the paste of the starch slowly with stirring into the boiling water fig. 10.1
4. Continue boiling for about 2-3 minutes and allow the beaker to cool.
5. Filter the contents of beaker through a filter paper fixed in a funnel fig. 10.2.

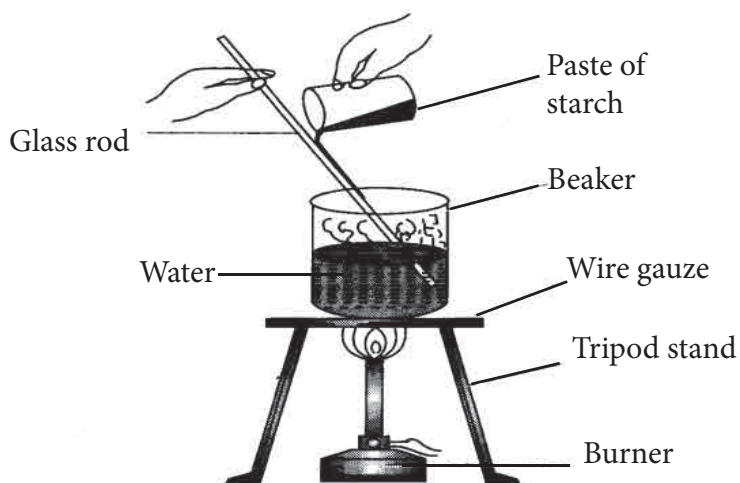


Fig. 10.1 Pouring of paste of starch in boiling water

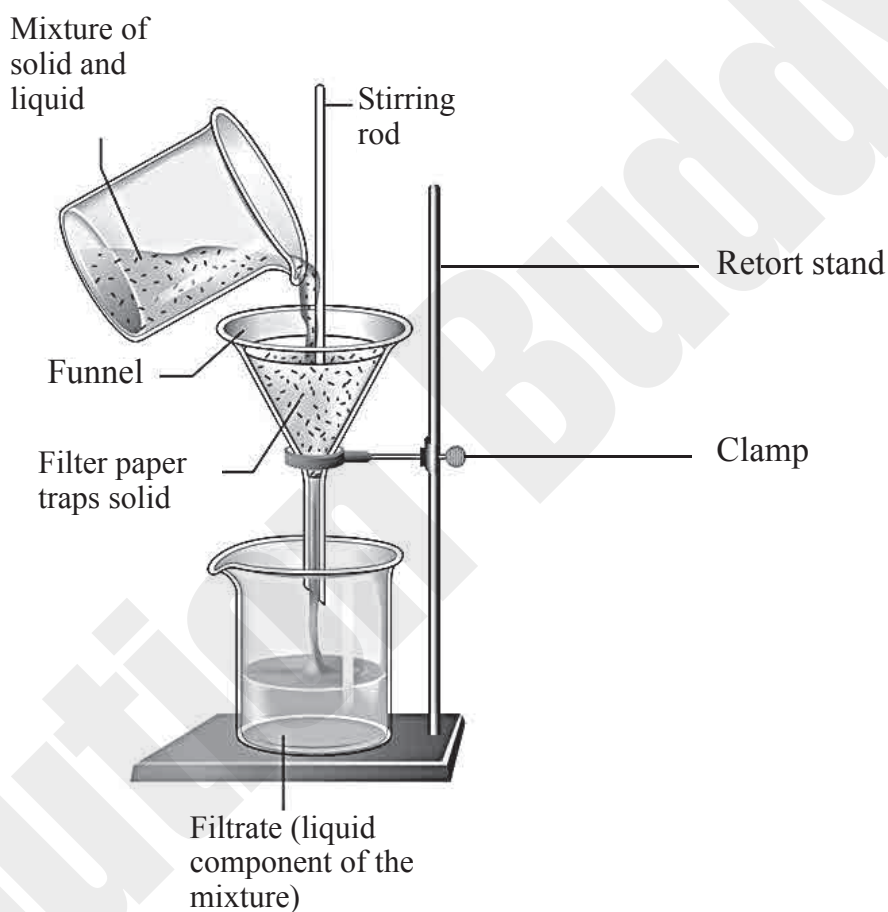


Fig. 10.2 Filtration

Observations:

Starch forms opaque / transparent sol easily and filtered rapidly through filter paper.

Conclusion:

A starch sol is prepared successfully. It is a lyophilic colloid, passes easily through filter paper, and shows strong affinity between the dispersed phase and the dispersion medium.

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. In the lyophilic colloids affinity between the particles of dispersed phase and dispersion medium is
a. strong ✓ b. high c. low d. average
2. The lyophilic sol is a
a. irreversible sol b. reversible sol ✓
c. hydrophobic sol d. homogeneous colloid
3. In a sol of starch, dispersion medium is
a. starch b. water ✓ c. ethyl alcohol d. air
4. The colloid of starch in water is
a. solid sol b. emulsion c. gel d. sol ✓
5. Which of the following is NOT true about colloidal system?
a. It is heterogeneous
b. Aggregation of molecules or ions ✓
c. The particles are settle down under gravity
d. The particles are invisible under microscope
6. Which one of the following is an example of sol?
a. Butter b. Jellies c. Milk d. Gelatin in water ✓
7. Which one of the following is an example of lyophilic sol?
a. $\text{Al}(\text{OH})_3$ sol b. $\text{Fe}(\text{OH})_3$ sol c. As_2S_3 sol d. Starch sol ✓

Short answer questions

1. What is a sol?

Ans.

A sol is a colloidal system in which a solid is dispersed in a liquid medium.

2. How sols are classified?

Ans.

Sols are classified into two types: lyophilic sols and lyophobic sols, based on the affinity between the dispersed phase and the dispersion medium.

3. What is lyophilic sol?

Ans.

A lyophilic sol is a colloidal solution in which the dispersed phase has a strong affinity for the dispersion medium.

4. Give two examples of lyophilic sols.

Ans.

Starch sol and gum sol.

5. Why lyophilic sols are called reversible sols?

Ans.

Lyophilic sols are called reversible sols because after evaporation of the dispersion medium, the sol can be reformed by simply adding the medium again.

Remark and sign of teacher:

NOTES /CALCULATIONS /ROUGH WORK